

From Data to Strategy: Building Operating intelligence for Qantas

Executive Proposal

Organisational priority and context

Qantas operates in a challenging aviation environment where operational performance plays an important role in customer satisfaction and safeguarding the reputation. When it comes to airline reliability, on time performance plays the most significant role as disruption such as cancellation and delays can influence customer satisfaction and operational flow (IATA 2024; Kim & Park 2016).

The main aim of this analysis is to improve operational reliability by understanding the factors that influence on-time performance, delays, cancellations and route performance across Australian airline industry.

Qantas is evaluated along with its key competitors such as Jetstar and Virgin Australia in order to better understand their performance. This view is quite significant as it provides an insight of its internal performance and airline's competitive standing. Delay can result in a series of disruptions across network such as disputed schedules, lower service reliability and inefficient ground crew and resources. Routes with large number of passengers play a much important role in network performance through punctuality, system-wide reliability and disruption patterns.

The recent disruption caused during the covid-19 pandemic highlighted the need of robust and reliable operational system within the airline industry. As the demand recovers, the executives rely on analytical insights to understand the reliability trends, areas of operational challenges and also evaluate the performance with other competitors. Therefore, the main aim of this proposal is to provide a relevant and evidence-based information that helps to assess operational reliability in the Australian aviation sector.

Evolution of the priority

The organizational priority has evolved significantly from Assignment 1 and 2. Initially, the main aim was to analyse Qantas through various metrics such as on-time performance, delay pattern and cancellation rates to identify patterns and potential problem areas.

As the project advanced, the aim of the project moved from initial exploration to execute decision support. Instead of analyzing operational issues one-by-one, the analysis is now focused on inclusive patterns that is highly relevant to operational performance. Several priorities have been combined and some have been removed in order to highlight the important metrics in the dashboard.

The final scope of the project has also been enhanced by shifting our focus from only Qantas to beyond Qantas. The final dashboard displays Qantas along with major competitors such as Jetstar

and Virgin Australia in order to understand the performance within the aviation market. This is quite significant in execute decision-making as most of the decision depend on relative market position. It also consists of three other visuals such as reliability performance on high-volume routes, the relation between delay and cancellation and seasonal delay patterns. These refinements provide meaningful insights that would support strategic planning and decision making.

Decision-making opportunity and Strategic Relevance

This analysis mainly focuses on evaluating operational reliability within the aviation market. Reliable operations are quite significant as it can impact customer satisfaction, brand reputation and operational efficiency. By properly identifying reliability issues and comparing them with industry standards helps the organisation to plan accordingly and allocate its resources effectively.

The analysis helps the executives to understand the reliability trend which makes it easier to determine whether the operation outcome is moving in a positive or negative direction. By comparing Qantas with other carriers ensures that reliability is analysed with the market standards and not just internal results.

Moreover, the analysis reveals the relation between network activity and operational performance by evaluating reliability in high flight volume routes. This highlights where operation disruptions can create a huge effect due to large number of passengers and the importance of those routes within the network.

By incorporating the analysis between delay and cancellation helps to understand where reliability issues are most common and where attention is most needed.

Seasonal trends are incorporated which helps to understand when reliability issue is usually at its peak across the industry so that the organization can consider operational planning and resource allocation accordingly.

The combined analysis helps to interpret the airline's operational effectiveness instead of focusing on individual operational issues.

Dashboard Concept and Rationale

The dashboard provides an overview of reliability, operational pressure and market comparison using a set of targeted visualisations. The visualisations are laid out to guide the user from the big-picture view to key factors that drive reliability.

The top-left graph helps to provide a quick overview of performance pattern over the years from 2004 to 2026 and how Qantas stack up against its key competitors. It provides an overall context for executive decision-making.

The top-right graph displays the top 10 high-volume Qantas routes using the data from the recent past year 2025 which highlights the routes where flight activity is high. It helps to identify areas where operational performance can have a huge effect on passengers and overall network performance.

The lower-left graph compares delay and cancellation rates which helps decision makers to understand how these metrics interact and helps to identify areas where operational disruption can impact the most.

The lower-right graph displays seasonal delay pattern in the 2025 which provides the current delay pattern across the year and helps the decision-makers to identify periods where operational pressure may intensify.

Simple visual design, clear title and layout has been used to easily interpret the data and a small note has been added to explain the different time periods used in the analysis in order to ensure transparency. Overall, the dashboard provides a quick insight and deeper operational analysis for executive decision-making.

Evidence Base

The analysis is based on Australian Domestic Airline On-Time Performance dataset by Bureau of Infrastructure and Transport Research Economics (BITRE). The dataset consists of on-time performance, delays, cancellations, flight volume and routes of Australian airlines.

Operational reliability is an important factor in aviation industry as disruption can have a negative impact on customer satisfaction, airline reputation and operational efficiency (IATA 2024; Kim & Park 2016). Metrics such as delay rates, cancellation rates and on-time performance are most commonly used for organizational monitoring and performance evaluation.

The dataset consists of data over the period of 2004 to 2026 which helps to analyse both the past and current reliability patterns. Historical trends help to understand how reliability has changed over time while 2025 route-level and seasonal data helps to capture most recent shift in route performance and seasonal delay patterns.

Incorporating the past and most recent data into the dashboard act as the strong evidence base for understanding reliability, competitive positioning and areas where there is a chance of potential operational pressure.

Decision Insights from the dashboard

Insight 1: Operational reliability Trend and competitive position

The dashboard displays how operational reliability has changed over the years and their performance when compared to major Australian airlines. This insight is quite important as reliability performance influences customer satisfaction and operational efficiency. This helps the

executives to understand the reliability trend across the aviation industry and also supports deeper consideration of operational performance and competitiveness.

Insight 2: Performance across high volume routes

The dashboard displays operational performance across high volume Qantas routes in 2025. As these routes play a significant role in network activity, they are important in providing insights on how reliability issues can affect customers and operational flow. This helps executives in considering network priorities and also operational performance.

Insight 3: Reliability Metrics and Operational Disruptions

The dashboard displays a different perspective of operational reliability through delay and cancellation rate. By analyzing these metrics helps to understand how disruption is spread across various reliability metrics. This insight provides a better understanding of service reliability and operational risk.

Insight 4: Seasonal Variation in Operational Performance

The dashboard displays the most recent change in delay rates across the months in the year 2025. Understanding this pattern is important when it comes to assessing the operational performance and understanding the market dynamics. This insight helps in considering resource allocation, planning priorities and taking necessary action to meet the expected operational performance throughout the year.

Dashboard Commentary and Analysis

Dashboard Design and Key Insights

In order to support executive-level decision-making, the dashboard mainly focused on selective group of charts instead of having a large number of exploratory charts. The dashboard is designed in a way that guides the user from overall performance trends to operation impacts and risk considerations followed by seasonal patterns that helps to analyse the future performance. Each graph is selected to contribute different aspects of airline reliability and keep information-focused to avoid unnecessary complexity. It helps the executives to understand the key issue, what factors are affected the most and what factors might affect the future result.

The first graph, *Operational Reliability Trend (2004-2026)* displays the pattern in on-time performance for major Australian airline over the past year. This helps the executive to understand the long-term trend and also helps to identify the periods where disruption has occurred. It also helps to analyse the performance of Qantas with its major competitors.

The second graph, *High-Volume Qantas Routes and Operational Impact (2025)* mainly focus on routes that has the highest number of passenger volume. By combining flight volume with reliability performance helps to identify which disruption issues can exert pressure on the organisation due to the number of services involved.

The third graph, *Reliability Risk: Delays and Cancellations* reveals the relation between delay and cancellation rates. This insight helps to understand the reliability risk and helps to analyse whether delays influence cancellation rates across the network.

The fourth graph, *Seasonal Pattern in Airline Delays (2025)* reveals the monthly delay rate among major Australian Airlines. This analysis helps to detect for any operational pressure and also helps to understand if operational pressures may be higher.

All these visuals together provide an overview of reliability, operational risk and seasonal patterns that shapes the overall performance of the airline.

How Dashboard Supports Strategic Decision-Making

The dashboard is designed with the intension of providing insight to support executive decision-making. Each graph contributes in understanding reliability and also different factors that play an important role in decision making.

The trend analysis helps to determine whether the current performance represent a short-term variation or is it part of a long-term pattern. This provides a strong ground for discussing about evaluating operational performance and shaping strategic priorities.

The route performance analysis draws attention to areas where operation disruption can cause the greatest business impact. This view helps in assessing reliability and service performance as the high-volume routes represent larger part of the network.

The delay and cancellation analysis draw attention to operational risk by displaying the interaction between two reliability measures. This view helps to assess the consequences that has to be faced due to weak operational performance.

The seasonal pattern analysis draw attention to operational challenges faced throughout the year. This view helps to identify the times of the year where their can be an increase in delay rate and additional resource allocation and operational readiness that might be needed.

Therefore, the dashboard is designed to support executive decision-making by displaying only the most relevant information that needs a close attention in a structured format.

Assumptions, Limitations and Future Improvements

The analysis assumes that comparing Qantas with Virgin Australia and Jetstar offers a relevant competitive assessment in aviation market. The most recent 2025 data is used in route and seasonal analysis as it is assumed that the most recent performance can contribute the most in decision making, while historical trend view is assumed to provide more information about the long-term patterns to understand the current operational challenges and risks.

The dataset is aggregated in month-route or airline level which restricts to analyse individual flight restriction or any specific events. Therefore, this limits from analysing the day to day variation.

The dataset also does not include information about passenger complaints, missed connections etc. due to which operational reliability is evaluated based on available indicators such as delay rate , on-time performance and cancellation rates which are most commonly used measures to understand airline service reliability.

Future improvements include incorporating real-time operational data, weather data and airport-level information. In order to predict reliability risk in the future, forecasting techniques can be included. By integrating the dashboard to a live operational system would help the executives to work with most recent information. With these improvements the dashboard would evolve from reviewing past performance to future focused insights.

Ethical and Privacy Considerations

The dashboard is created using publicly available dataset which is published by Bureau of Infrastructure and Transport Research Economics (BITRE). This dataset consists of information about airline performance and does not include any personal information of any passenger or employees. Therefore, due to the absence of personal data, privacy risk is very minimal. Nonetheless, responsible use of dashboard is quite significant. Reliability measures must be contextualized and it should not be used to draw conclusion on organizational performance without considering other factors.

To use the dashboard responsibly, transparency plays a very important role. The data source that has been used is included in Tableau workbook in order to present the source of the information.

If the future version of the dashboard includes live operational or customer data, more privacy and security have to be incorporated. Ensuring that the information is accurate and well-protected is important in confident decision-making.

REFERENCES

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